PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Global epidemiology of COVID-19 knowledge, attitude and practice - a systematic review and meta-analysis
AUTHORS	Siddiquea, Bodrun; Shetty, Aishwarya; Bhattacharya, Oashe; Afroz, Afsana; Billah, Baki

VERSION 1 – REVIEW

REVIEWER	Abushouk, Abdelrahman Ain Shams University
REVIEW RETURNED	08-Apr-2021

GENERAL COMMENTS	The authors exerted significant efforts to synthesize evidence on KAP towards COVID-19 around the world. While their search and analysis methods have been thorough, I have some comments: - This topic is very hot with few studies published every single week. Stopping the search in December 2020 does not reflect the current KAP, especially after the events in the subsequent months as the introduction of vaccines and appearance of new variants. The rapid publications on COVID-19 made some experts call for live systematic reviews that are updated instantly once relevant publications are added. I am aware of several studies that could have been included, especially if the search was updated to reflect current KAP state. Our group has recently published a study of 72,000 participants in 22 countries (https://bmjopen.bmj.com/content/11/2/e043971). In a supplementary file, we summarized > 20 relevant studies that the authors can look at. Further, I would advise the authors to look at the >1000 papers citing Zhong et al. (https://www.ijbs.com/v16p1745.htm) which is one of the earliest papers in this regard). Many of these papers have replicated Zhong et al. study all over the world. - One of the major limitations in conducting such analysis is the wide variability in the used questionnaires, timing and place of data collection, setting of participants, etc. One important variable (timing) may require further analysis (beyond just correlation with the practices score).One possible approach is to divide studies by the time of data collection (not publication) into 3 months intervals or according to important events along the course of the pandemic e.g. WHO pandemic declaration, First vaccine approval, etc. - It would significantly improve the manuscript if the authors use better data visualization techniques. For example, comparison of countries KAP can be placed on a map.

	Specific Comments * Abstract - What does the authors mean when they say the pooled proportion for knowledge for example was 74%? That a participating individual was likely to answer around 74% of the knowledge questions right or 74% of participants in a single study was 74%? - The results section of the abstract needs more details and supporting statistics. - "The pooled proportions of KAP components were ...." * Strengths and limitations - "The methodological design and statistical analyses are sound and robust". I honestly would recommend leaving such judgement to the reader. - Other limitations such as heterogeneity and language restrictions should be briefly mentioned as well. * Introduction - Up to 32% may require intensive care? The authors should add the context for this number as it is quite high. Are these patients with specific comorbidities? - The study would be much stronger and have a practical value if it updated its search to include recent studies that assessed knowledge and attitudes about vaccines. * Methods - As this is a meta-analysis of observational studies, the MOOSE checklist may be a better fit. - Page 4, Line 35: which preprint servers did the authors search? - I am not sure of the method the authors used to conduct subgroup analysis. I see you pooled studies with shared qualities together, but you report the numbers separately (for example <30 years vs older). Did you use a comparative test or just visually compared the numbers as the latter methods doesn't mean the differences are of statistical value. - In the presence of a publication bias, did the authors use any method as trim and fill to correct for it? - A comparison by continent isn't valuable in my opinion. I would say a comparison by country would inform governments better. A map would be valuable in that regard as mentioned earlier. * Discussion - "The pooled score of KAP was moderate". What does this mean? Did the authors pool knowledge, attitudes, and practices scores altogether? - " This could be due to females' innate fear for their children and family which drive them to adopt the preventive behaviors or males' earning role in the family which make them to take risks of going out". This explanation would be controversial and may not fully reflect reality in the modern world. If the authors decided to keep it, I would advise using supporting references. - The discussion is generally well-written.

REVIEWER	Balduzzi, Sara University of Freiburg, Institute of Medical Biometry and Statistics
REVIEW RETURNED	19-Apr-2021

GENERAL COMMENTS	The authors conducted a systematic review to assess the knowledge, attitude and practice regarding COVID-19 in the general population. The methods are appropriate and clearly described, and the results are well presented. I have only a couple of comments: *** Introduction I would suggest to use the extended of "KAP" before using the acronym. *** Methods: Search strategy, information sources and study selection PRISMA instead of PRISMA-P (Table 1 reports the PRISMA Checklist), and related citation. *** Methods: Data Analysis - "I2 values of 0–25% show low heterogeneity, 26–75% show moderate heterogeneity and 76–100% show substantial heterogeneity." The reference for this statement is the first published to describe I-squared. Nowadays, overlapping ranges to interpret it are usually prefered (see the Cochrane Handbook, par 10.10.2 Identifying and measuring heterogeneity, https://training.cochrane.org/handbook/current/chapter-10#section-10-10-2). In the context of this review, it might be reasonable to simply state "I2 values over 75% might represent substantial/considerable heterogeneity" - I would add a line to better describe the correlation analysis (and the use of r)

REVIEWER	Costa-Santos, Cristina Faculty of Medicine, University of Porto, Centre for Research in Health Technologies and Information Sy stems (CINTESIS), Information Sciences and Decision on Health Departm ent (CIDES)
REVIEW RETURNED	14-May-2021

GENERAL COMMENTS	The authors performed a systematic review and meta-analysis in order to provide the current global epidemiology of COVID-19 knowledge, attitude and practice in the general population. However, the fact that the authors are unable to identify the source of the huge heterogeneity presented is the main weakness of this study. Comments to the Author: 1. Only published articles that had full text available in English were included, however excluding languages other than English may introduce a bias and lead to erroneous conclusions. Authors should discuss this limitation. 2. Two researchers independently conducted the screening of the title and abstract for each article obtained from the literature search. After the independent review, any disagreement on article inclusion was resolved by discussion or by consulting with a senior researcher. The results on agreement between the two researchers should be presented in the manuscript. What was the percentage of agreement between the two researchers? Considering the articles in which there was no agreement between

	the researchers, how many were resolved by consensus and how many were resolved with a third reviewer?

REVIEWER	Quinn, Megan East Tennessee State University College of Public and Allied Health, Biostatistics and Epidemiology
REVIEW RETURNED	14-May-2021

GENERAL COMMENTS	Thank you for the opportunity to review this manuscript. The work is timely and worthwhile, providing a comprehensive review and analysis that is needed. Some suggestions for revision: Introduction: suggestion to add some information about why KAP in China might be different than other countries since this is part of the analysis. Though this may be obvious currently in the pandemic, this may not be obvious if someone were to read this in the future. This information could be added in the methods instead, if that's more appropriate. Page 4, Line 14: suggestion to change "it" to COVID-19 or the pandemic to specify what "it" is referring to. Page 7, Line 22: literature likely does not need to be plural Page 8, Lines 4-29: suggestion to modify to make it a bit more clear on what items were significantly different from each other. As written it is not entirely clear unless the reader understands how to determine significance from the confidence intervals. Page 8,Line 10: add a % sign after 90 to be consistent with over CI reporting Page 9, Line 26-28: please check the % signs for the CIs and add percents where needed to be consistent with other CI reporting. General comment: Check where the decimal point is located and ensure consistency throughout the manuscript. Sometimes the decimal is centered vertically and other times it is at the bottom.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Abdelrahman Abushouk, Ain Shams University
Comments to the Author:
The authors exerted significant efforts to synthesize evidence on KAP towards COVID-19 around the world. While their search and analysis methods have been thorough, I have some comments:

4. Comment:- This topic is very hot with few studies published every single week. Stopping the search in December 2020 does not reflect the current KAP, especially after the events in the subsequent months as the introduction of vaccines and appearance of new variants. The rapid publications on COVID-19 made some experts call for live systematic reviews that are updated instantly once relevant publications are added.

I am aware of several studies that could have been included, especially if the search was updated to reflect current KAP state. Our group has recently published a study of 72,000 participants in 22 countries (https://bmjopen.bmj.com/content/11/2/e043971). In a supplementary file, we summarized > 20 relevant studies that the authors can look at. Further, I would advise the authors to look at the >1000 papers citing Zhong et al. (https://www.ijbs.com/v16p1745.htm) which is one of the earliest

papers in this regard). Many of these papers have replicated Zhong et al. study all over the world.

Response: The search has been updated to the present time (1 January 2020 to 30 June 2021) and the recommended/relevant articles have been added to this revised manuscript.

5. Comment: One **of the major limitations** in conducting such analysis is the wide variability in the used questionnaires, timing and place of data collection, setting of participants, etc. One important variable (timing) may require further analysis (beyond just correlation with the practices score).One possible approach is to divide studies by the time of data collection (not publication) into 3 months intervals or according to important events along the course of the pandemic e.g. WHO pandemic declaration, First vaccine approval, etc.

Response: Further analysis has been done by time of data collection (study period) into three months interval and the results have been added to the revised manuscript. However, the heterogeneity has not been improved (Lines 363−368, results section, page 8; lines 484−494, discussion section, page 10).

6. Comment: It would significantly improve the manuscript if the authors use better data visualization techniques. For example**, comparison of countries KAP can be placed on a map.**

Response: We have added comparison of countries KAP on a map as suggested (Figure 3).

Specific Comments

*** Abstract**

7. Comment - What does the authors mean when they say the pooled proportion for knowledge for example was 74%? That a participating individual was likely to answer around 74% of the knowledge questions right or 74% of participants in a single study was 74%?

Response: Sorry for the lack of clarity in the presentation of the statement. By pooled proportion for knowledge we mean the proportion of people in the present study that answered correctly in the knowledge questions. The comment has been addressed and the statement has been re-framed for clarity.

The following text has been added in the 'abstract' section; Lines 67−69, page 2.

'The estimated overall correct answers for knowledge, good attitude and good practice in this review were 75% (95% CI 72−77%), 74% (95% CI 71−77%) and 70% (95% CI 66−74%) respectively.'

The following text has been added in the 'results' section; Lines 321−326, page 7.

'Figure 2 represents the pooled proportion of people who had answered correctly in their knowledge questions, had positive attitude and good practice techniques. The findings demonstrated that 75% of the present study population answered the knowledge question correctly (95% CI 72–77%, $p < 0.001$, $I^2 = 99.55\%$), 74% had positive attitude towards COVID-19 (95% CI 71–77%, $p < 0.001$, $I^2 = 99.58\%$) and 70% followed good COVID-19 practice techniques (95% CI 66–74%, $p < 0.001$, $I^2 = 99.69\%$).'

The following text has been added in the 'discussion' section; Lines 446−447, page 10.

‘The estimated overall correct answers for knowledge, good attitude and good practice techniques in this review were 75%, 74% and 70% respectively.’

8. Comment - The results section of the abstract needs more details and supporting statistics.

Response: The comment has been addressed partially. However, because of word limitation in the abstract, sorry that all supporting statistics could not be added.

9. Comment - "The pooled proportions of KAP components were ...."

Response: This has been revised; please see response of Comment 7 above.

* **Strengths and limitations**
10. Comment: "The methodological design and statistical analyses are sound and robust". I honestly would recommend leaving such judgement to the reader.

Response: This comment has been addressed and the line has been deleted.

11. Comment: Other limitations such as heterogeneity and language restrictions should be briefly mentioned as well.

Response: The limitation of heterogeneity and that of language restrictions has been discussed in the discussion section in the revised manuscript (lines 546-549, page 12). However, due to limitations in the number of points that could be included in the strengths and limitations section, they were excluded from the original manuscript. We have now added the following limitations: Lines 95−96, page 3.

‘Articles in non-English languages were excluded, and substantial heterogeneity was present in the evaluated outcomes.’

***Introduction**
12. Comment: Up to **32%** may require intensive care? The authors should add the context for this number as it is quite high. Are these patients with specific comorbidities?

Response: The comment has been addressed by replacing the existing text with the following text, extremely sorry that there was a typo in the earlier reported % (32%); lines 110−112; page 3.

‘and it mainly affects the respiratory system and can lead to acute hypoxaemic respiratory failure, and between 0.9% to 14% of these patients require admission to intensive care units (ICU) for advanced respiratory support.’

13. Comment: The study would be much stronger and have a practical value if it updated its search to include recent studies that assessed knowledge and attitudes about vaccines.

Response: Thank you for the comment. Knowledge and attitudes about vaccines is indeed a very interesting area that needs significant research. However, we believe that the topic is very vast and would require a systematic review and meta-analysis of its own. Besides, the aim of the study from the very beginning did not take into account the knowledge and attitude towards vaccines. Hence, to stay in line with the original aim of the study, the search strategy was not updated. However, we have acknowledged the fact that until the various barriers in vaccine roll-out do not let the world reach its

level of herd immunity, there is a need to rely on enhancing people's KAP on COVID-19. This is present in the following lines 130−132, page 3 in the introduction section.

'and a recent global systematic review on COVID-19 vaccine hesitancy among general population showed that in many countries vaccine acceptance is below 60%.'

***Methods**
14. Comment: As this is a meta-analysis of observational studies, the MOOSE checklist may be a better fit.

Response: In addition to PRISMA checklist, MOOSE checklist has been added (Supplementary Table 2).

15. Comment: Page 4, Line 35: which preprint servers did the authors search?

Response: Any preprint servers had not been searched, but we included the preprint found by manual search.

16. Comment: I am not sure of the method the authors used to conduct subgroup analysis. **I see you pooled studies with shared qualities together**, but you report the numbers separately (for example <30 years vs older). Did you use a comparative test or just visually compared the numbers as the latter methods doesn't mean the differences are of statistical value.

Response: Pooled proportion and their 95% confidence interval were used to compare the groups. For example: Participants aged 30 years and above had relatively higher knowledge (80%, 95% CI 75−85% vs 78%, 95% CI 74−83%), attitude (73%, 95% CI 65−81% vs 68%, 95% CI 59−78%) and practice (82%, 95% CI 74−90% vs 80%, 95% CI 72−88%) compared to those aged below 30. However, the difference was not significant as the respective confidence intervals overlapped.

Thus, this comment has been addressed by adding the following text in the 'results' section (lines 341−345, page 7−8). Furthermore, as suggested in the 'Comment 32', a part of the results section has been edited for better clarity.

'Respondents aged 30 years and above had relatively higher knowledge (80%, 95% CI 75−85% vs 78%, 95% CI 74−83%), attitude (73%, 95% CI 65−81% vs 68%, 95% CI 59−78%) and practice (82%, 95% CI 74−90% vs 80%, 95% CI 72−88%) compared to those aged below 30. However, the difference was not significant as the respective confidence intervals overlapped.'

17. Comment: In the presence of a publication bias, did the authors **use any method as trim** and fill to correct for it?

Response: Sorry, we did not use the 'trim and fill in' method to correct publication bias.

18. Comment: A comparison by continent isn't valuable in my opinion. **I would say a comparison by country would inform governments better.** A map would be valuable in that regard as mentioned earlier.

Response: The comment has been addressed by adding a map with colour gradient (from light to dark) in every map representing country scores (from low to high), please see figure 3: Geographical representation of knowledge, attitude and practice during the COVID-19 pandemic.

*** Discussion**

19. Comment: "The pooled **score of KAP was moderate". What does this mean? Did the authors pool knowledge, attitudes, and practices scores altogether**?

Response: Sorry for the lack of clarity in the statement. The pooled score referred to the pooling of scores for the individual components of knowledge, attitude and practice. The line has been reframed in the discussion section (line number 440−441, page 9).

'The study demonstrated a pooled score of ≥70% in each of the three components of KAP.'

20. Comment: " This could be due to females' innate fear for their children and family which drive them to adopt the preventive behaviors or males' earning role in the family which make them to take risks of going out". This explanation would be controversial and may not fully reflect reality in the modern world. If the authors decided to keep it, I would advise using supporting references.

Response: This comment has been addressed by deleting these lines.

21. Comment: The discussion is generally well-written.

Response: Thank you.

Reviewer: 2
Dr. Sara Balduzzi, University of Freiburg
Comments to the Author:
+++The authors conducted a systematic review to assess the knowledge, attitude and practice regarding COVID-19 in the general population. The methods are appropriate and clearly described, and the results are well presented.

I have only a couple of comments:

*** **Introduction**

22. Comment: I would suggest to use the extended of "KAP" before using the acronym.

Response: This has been addressed by replacing 'KAP' with 'knowledge, attitude and practice (KAP)'. Line 163, introduction section, page 4.

*** **Methods:**

23. Comment - Search strategy, information sources and study selection
PRISMA instead of PRISMA-P (Table 1 reports the PRISMA Checklist), and related citation.

Response: 'PRISMA-P' has been replaced with 'PRISMA' and has been cited as well (reference 15). Line 200, methods section, page 5 and supplementary page 2.

*** **Methods: Data Analysis**

24. Comment:
- "I2 values of 0–25% show low heterogeneity, 26–75% show moderate heterogeneity and 76–100% show substantial heterogeneity." The reference for this statement is the first published to describe I-squared. Nowadays, overlapping ranges to interpret it are usually prefered (see the Cochrane Handbook, par 10.10.2 Identifying and measuring heterogeneity, https://training.cochrane.org/handbook/current/chapter-10#section-10-10-2). In the

context of this review, it might be reasonable to simply state "I2 values over 75% might represent substantial/considerable heterogeneity"

Response: The comment has been addressed in the ‘data analysis’ part of ‘method’ section. Lines 280−281, page 6.

25. Comment: - I would add a line to better describe the correlation analysis (and the use of r)

Response: The following text has been added in the ‘data analysis’ part of ‘methods’ section. Lines 293−294, page 7.

‘to evaluate the strength and direction of relationship (denoted by r) between knowledge and attitude, attitude and practice, and knowledge and practice.’

Reviewer: 3
Dr. Cristina Costa-Santos, Faculty of Medicine, University of Porto
26. Comments to the Author:
The authors performed a systematic review and meta-analysis in order to provide the current global epidemiology of COVID-19 knowledge, attitude and practice in the general population. However, the fact that the authors are unable to **identify the source of the huge heterogeneity presented is the main weakness of this study.**

Response: A comprehensive analysis was performed to identify the source of heterogeneity without success. This limitation has been acknowledged in the ‘strengths and limitations’ (lines 95−96, page 3) as well as ‘discussion’ section (lines 546−548, page 12). Please see response of Comment 11 above.

Comments to the Author:

27. Comment: Only published articles that had full text available in English were included, however excluding languages other than English may introduce a bias and lead to erroneous conclusions. Authors should discuss this limitation.

Response: This has been acknowledged in the ‘strengths and limitations’ section (line 95, page 3). Please see response of Comment 11 above.

28. Comment: Two researchers independently conducted the screening of the title and abstract for each article obtained from the literature search. After the independent review, any disagreement on article inclusion was resolved by discussion or by consulting with a senior researcher. The results on agreement between the two researchers should be presented in the manuscript. What was the percentage of agreement between the two researchers? Considering the articles in which there was no agreement between the researchers, how many were resolved by consensus and how many were resolved with a third reviewer?

Response: The comment has been addressed and the following text has been added in ‘methods’ section. Line 211−213, page 5.

‘They had a disagreement with the consideration of 4 papers (5%).Three papers were resolved by mutual consensus and one was resolved by consulting with the senior author (BB).’

Reviewer: 4
Dr. Megan Quinn, East Tennessee State University College of Public and Allied Health
Comments to the Author:
Thank you for the opportunity to review this manuscript. The work is timely and worthwhile, providing a comprehensive review and analysis that is needed.

Some suggestions for revision:
29. Comment: Introduction: suggestion to add some information about why KAP in China might be different than other countries since this is part of the analysis. Though this may be obvious currently in the pandemic, this may not be obvious if someone were to read this in the future. This information could be added in the methods instead, if that's more appropriate.

Response: The following text has been added in the 'methods' section in page 6, lines 286−289.

'A comparison of China vs other countries was done because China was the epicentre of COVID-19, and was the first country to introduce the preventive measures and strict laws. Hence, we wanted to analyse if this difference in timeline would have an effect on its people's knowledge, attitude and practice in comparison with other countries.'

30. Comment: Page 4, Line 14: suggestion to change "it" to COVID-19 or the pandemic to specify what "it" is referring to.

Response: The word 'it' has been replaced with 'COVID-19' (page 3, line 109, introduction section).

31. Comment: Page 7, Line 22: literature likely does not need to be plural

Response: 'literatures' has been replaced with 'literature' (page 7, line 297, patient and public involvement section).

32. Comment: Page 8, Lines 4-29: suggestion to modify to make it a bit more clear on what items were significantly different from each other. As written it is not entirely clear unless the reader understands how to determine significance from the confidence intervals.

Response: The text has been reviewed and updated as suggested. It is now lines 341−370, page 7−8 in the revised manuscript.

'Respondents aged 30 years and above had relatively higher knowledge (80%, 95% CI 75−85% vs 78%, 95% CI 74−83%), attitude (73%, 95% CI 65−81% vs 68%, 95% CI 59−78%) and practice (82%, 95% CI 74−90% vs 80%, 95% CI 72−88%) compared to those aged below 30. However, the difference was not significant as the respective confidence intervals overlapped. There were no significant differences between male and female participants regarding prevalence of knowledge, attitude and practice, however practice scores were slightly lower among male participants (75%, 95% CI 69−80%) than females (77%, 95% CI 72−82%). Participants with 12 years of education or less had non-significant lower prevalence of knowledge and practice than, but similar attitudes to, those with above 12 years of education. There was no significant difference between employed and unemployed participants when compared by each component of KAP. Knowledge score ranged between 72% (95% CI 63−80%) for low income countries and 79% (95% CI 75−83%) for upper-middle income countries. There was no progressive increase or decrease in attitude and practice scores by income level of the countries. However, low income countries had the lowest attitude score (69%, 95% CI 62−76%) and that was the highest for upper-middle income countries (78%, 95% CI 72−84%). In the context of practicing scores, low-income countries had the lowest (55%, 95% CI 46−64%), and the highest came from the upper-middle income countries (77%, 95% CI 70−84%) and the difference was statistically significant. Analysis by continent showed non-significant differences. However, Americas had the highest knowledge score (79%, 95% CI 74−83%) and that was the lowest in Europe/Oceania (67%, 95% CI 54−80%). Attitude

scores were highest in Middle-east (79%, 95% CI 72−85%) and lowest in Americas (66%, 95% CI 56−77%). Asia had the non-significant highest practice score (76%, 95% CI 71−81%) as compared to other continents. Further subgroup analysis was done by time of data collection (study period) into three months interval (January-March 2020, April-June 2020, July-October, 2020). There was no significant increase or decrease in knowledge and practice over time. However, positive attitude towards COVID-19 had been decreased significantly over the time (January-March 2020: 80%, 95% CI 74−86%; April-June 2020: 73%, 95% CI 68−77%; July-October 2020: 65%, 95% CI 57−73%). Heterogeneity was high (I^2 > 98%) across all subgroup analyses (except for Europe/Oceania for attitude and practice where the heterogeneity was 0.29% and 0.28% respectively), and publication bias was present in most of the studies.'

33. Comment: Page 8, Line 10: add a % sign after 90 to be consistent with over CI reporting

Response: % sign has been added (page 7, line 343).

34. Comment: Page 9, Line 26-28: please check the % signs for the CIs and add percents where needed to be consistent with other CI reporting.

Response: Comment has been addressed (results section, page 7−8).

35. General comment: Check where the decimal point is located and ensure consistency throughout the manuscript. Sometimes the decimal is cantered vertically and other times it is at the bottom.

Response: The comment has been addressed thoroughly. The decimal point is located at the bottom throughout the manuscript.

****Note:** Based on the reviewer's comment, we have done subgroup analysis by study period of each component of KAP and the results have been added to the revised manuscript. So, we have excluded the subgroup analysis by publication date (figure 1d in the original supplementary file) from the revised supplementary file. However, if the editors deem that it is essential to include them, then we will do as advised.

VERSION 2 – REVIEW

REVIEWER	Abushouk, Abdelrahman Ain Shams University
REVIEW RETURNED	10-Jul-2021

GENERAL COMMENTS	The authors have addressed my earlier comments to the best of their ability.